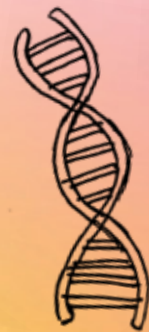


Shine

Fall 2025

The Power of Precision

Three forces that shape a child's health—and how experts are learning to read them all



**ALSO IN
THIS ISSUE**

Studying proteins to unlock tailored cancer therapies

When genes predict side effects before treatment begins

Why a child's lived experience holds the key to their health

**BC Children's
Hospital**
Foundation

- 02 The whole story
- 06 Beyond the blueprint
- 08 A first among 8 billion
- 09 Life's dimmer switch
- 10 The right match
- 12 The power of being seen
- 13 Micro moment

BOARD OF DIRECTORS

Mike Brankston, *Chair*
Martin Thibodeau, *Vice-Chair & Treasurer*

Sophia Alibhai
Jonathan Becker
Kelly Blankstein
Dale Boniface
Chris Chan
Vanessa Flockton
Howard Grant, O.B.C.
Jeff Hanman
Julia Kim, *Immediate Past Chair*
Pehlaj Malhotra
Catherine McLeod-Seltzer
Dr. Steven Miller
Will Westeringh, KC
Jessica Yan Macintosh

EMERITUS PAST CHAIR

Lisa Hudson (2022)
Don Lindsay, O.B.C. (2023)
Darrell Jones (2025)

FOUNDATION EXECUTIVE

Malcolm Berry
President & CEO

Matthew Boydston
Vice-President & Chief Financial Officer

Raquel de Munain
Vice-President, People & Culture

Christopher Melonie
Vice-President, Business Intelligence & Information Technology

Surina Sproul
Vice-President, Brand Strategy

Rita Thodos
Executive Vice-President, Philanthropy

Joanne Waxman
Vice-President, Strategic Partnerships

For more information about the editorial content of SHINE or to make a donation to BC Children's Hospital Foundation, please contact 604.449.6333, toll-free at 1.833.315.2133 or info@bcc hf.ca.

Charitable Business Number: 118852433RR0001

BC Children's Hospital Foundation
938 West 28th Avenue,
Vancouver, BC V5Z 4H4

Return undeliverable Canadian addresses to address above.

SHINE is published semi-annually by BC Children's Hospital Foundation.

Managing Editor: Ali Omelaniec

Writers: Lauren Erdelyi, Miriam Mortimer

Creative Director: Terry Lin

Graphic Design: Jennifer Pratt

@BCCHF   

A new lens on health



In just hours, health care providers can sequence a child's entire genome—six billion letters of DNA that would fill thousands of books. This extraordinary blueprint can reveal the origins of rare diseases and aggressive cancers. Yet genetics is only the beginning of a child's health story.

The rest unfolds in unexpected places. Trillions of microbes evolve daily within each child. Molecular switches respond to stress and environment, turning genes on and off like dimmer switches. Remarkably, postal codes can predict health outcomes as powerfully as genetic

codes. These forces shape each child's health in ways we're only beginning to understand.

This complexity explains why the same treatments produce radically different results. An asthma medication that saves one child may fail another whose microbiome tells a different story. The cancer therapy that is effective for one patient might cause heart complications in another. Even mental health emerges from this intricate web of biological and social influences, each thread essential to the whole.

At BC Children's Hospital, precision health means reading every dimension of a child's story—and these capabilities have never been stronger. Experts zoom in to examine the smallest proteins and zoom out to study entire neighbourhoods. They grow miniature replicas of conditions to test treatments before administering a single dose. They're even starting to correct genetic errors, letter by letter.

For the first time, we have the ability not just to understand each child's full complexity, but to help transform their health outcomes. Thank you for standing with us as we build this new era of personalized care, where every child's treatment is as unique as they are.

With gratitude,

Malcolm Berry
President & CEO
BC Children's Hospital Foundation

Inside each child, six billion letters of genetic code tell a powerful story. But it's just the beginning. DNA provides the foundation, yet accounts for only a fraction of what determines a child's health. The rest unfolds in surprising places: in trillions of microbes that train the developing immune system, in environments that switch genes on and off, and in postal codes that shape health as profoundly as genes do.

At BC Children's Hospital, researchers and clinicians are reading all these chapters at once. Because only the complete story can guide care that's truly personalized.

THE WHOLE STORY

Where biology, microbes and lived experience meet

MASTERING THE LANGUAGE OF DNA

Medicine's most complex puzzles often trace back to a child's genes. A single gene mutation can trigger a rare disease. A missing DNA segment might cause cystic fibrosis. Childhood cancers can emerge when multiple mutations align, setting off a reaction that traditional treatments struggle to stop. Chemotherapy and radiation take a broad-strokes approach that attacks healthy cells alongside cancerous ones, but don't target the mutation at the root of it all.

That approach is changing. At BC Children's, doctors can now sequence a child's entire healthy or cancer cell genome—scanning six billion DNA letters to pinpoint the cause or driver of their illness. What once took years and cost millions of dollars can

now happen in days, for less than the price of a smartphone.

This fall brings a major leap forward in precision cancer care. BC Children's is launching OPTIMISE, a Canadian first. When genetic testing reveals a targetable mutation in a cancer, OPTIMISE matches that child to therapies designed for their exact cancer profile. Unlike traditional trials that explore a single treatment, it uses an "adaptive design" that allows multiple targeted treatment arms to run at once—each tailored to specific genetic alterations found in the tumour.

"This is a major step forward because it allows us to connect kids with gene changes to treatments they would never otherwise access," said Dr. Rebecca Deyell, a pediatric oncologist and Mojgan Mowafaghian Chair

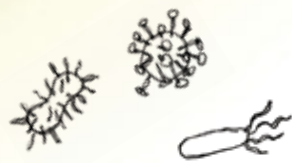


Shepherd, age 2, in treatment for leukemia
Also pictured on cover

Information courtesy of the Canadian Medical Association



L to R: Dr. Rebecca Deyell with Shepherd, age 2 and his mom, Karen



“We’ve neared the limits of what we can do with conventional therapy,” Dr. Li said. “The most important way we can improve outcomes—while decreasing side effects—is to do things more precisely.”

THE MICROBE FACTOR

As scientists master the language of DNA, they’re discovering genes don’t tell the whole story. Some of medicine’s most important clues now come from the trillions of microbes that influence how genes behave.

Consider asthma. One in five children develop it, yet traditional methods treat them all the same way. Dr. Stuart Turvey, a pediatric immunologist and investigator at BC Children’s Hospital and Canada Research Chair in Pediatric Precision Health, is leading research that reveals why this approach sometimes fails. His team found that the specific bacteria inhabiting a baby’s gut in the first months of life can determine whether asthma develops years later.

This breakthrough opens promising possibilities. By supplementing the exact missing microbes early in life—think of it as a probiotic prescription tailored to each child’s unique needs—doctors could potentially prevent asthma before the first wheeze.



“We’ve neared the limits of what we can do with conventional therapy. The most important way we can improve outcomes—while decreasing side effects—is to do things more precisely.”

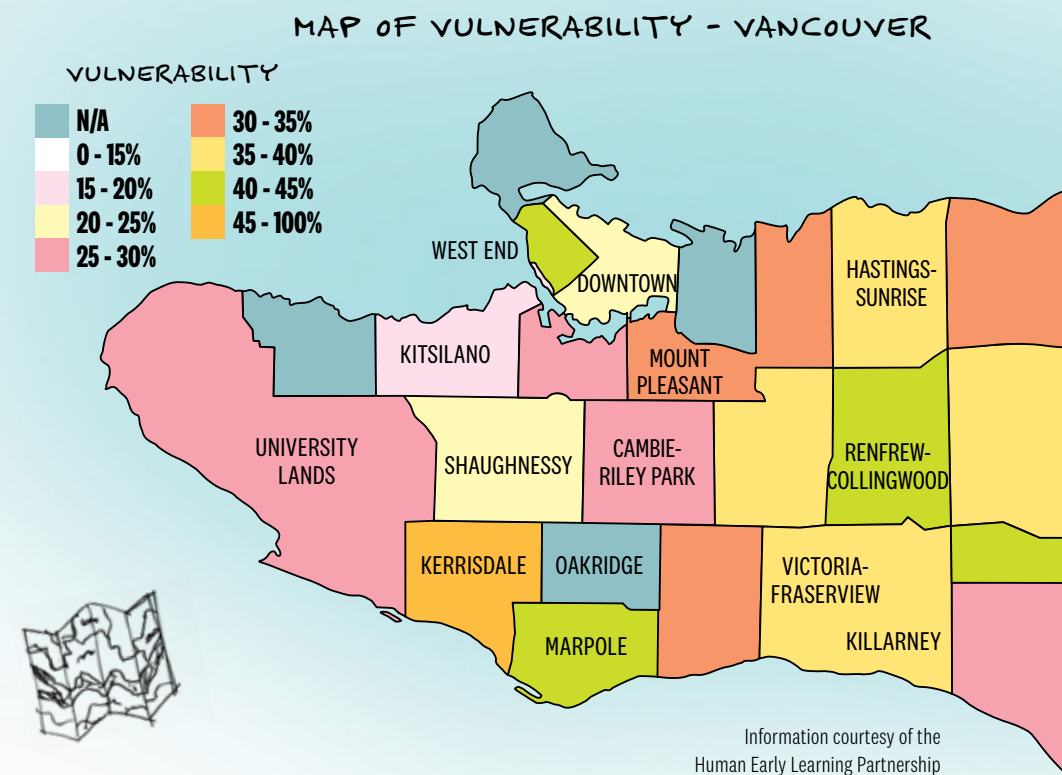
— Dr. Amanda Li

THE OTHER HALF OF HEALTH

Dr. Matthew Carwana, a pediatrician and researcher at BC Children’s, points to a map of Vancouver, its colourful patches representing different neighbourhoods. A child born in one area may face far greater health vulnerabilities than one born just 20 minutes away.

“Social and structural factors, which include income, housing stability, access to healthy food and systemic inequities, shape at least half of health outcomes,” he said. Among high-income countries, Canada ranks poorly for child well-being. The reason lies not in medical limitations but in deep structural inequalities.

In his upcoming research, Dr. Carwana hopes to use advanced MRI technology to improve health outcomes for children who have experienced early-life adversity. Chronic stress from poverty or discrimination can physically alter developing brains, change how genes express themselves and disrupt immune function.



Information courtesy of the Human Early Learning Partnership

Like a dimmer switch on DNA, lived experience can dial a child’s potential up or down.

“By bringing a health equity lens to precision medicine,” he said, “we can move beyond checklists and better understand how these kids’ brains are actually working in order to do a better job at predicting which therapies will be most effective.”

WHERE ALL THREADS MEET

True precision health emerges when these threads—genetics, microbes and lived experience—are woven together. For some children, one factor will dominate. A genetic mutation sometimes drives disease, or social adversity can overwhelm biology. But for most, health is determined by the dynamic interplay of all three.

At BC Children’s Hospital, this is already becoming reality as experts read every chapter of a child’s health story, from the genetic code they inherit, to the microbes they harbour, to the postal code they call home. As Dr. Steven Miller, Head of Pediatrics and Hudson Family Hospital Chair in Pediatric Medicine at BC Children’s Hospital emphasizes, “The real advances happen when we work at this intersection.”

It’s in that intersection that precision health finds its power. Only by understanding the whole story can we help write the endings every child deserves.★



L to R: Dr. Audi Setiadi, hematopathologist, and Dr. Amanda Li in the lab

BEYOND THE BLUEPRINT

THE HIDDEN STORY PROTEINS TELL

Dr. Philipp Lange with a mass spectrometer



“Technology is giving us the tools to see every patient as an individual, and to treat them that way.”

— Dr. Philipp Lange

IN HIS LABORATORY AT BC CHILDREN’S HOSPITAL, DR. PHILIPP LANGE WATCHES AS A **MASS SPECTROMETER** HUMS QUIETLY WHILE PERFORMING WHAT SEEMED IMPOSSIBLE JUST YEARS AGO. FROM A TINY PIECE OF TUMOUR, IT DECODES THOUSANDS OF PROTEINS IN MINUTES TO REVEAL WHAT’S HAPPENING INSIDE A CHILD’S CANCER.

The technology bridges a gap that has frustrated doctors for decades: the difference between knowing what genes say can happen and seeing what bodies are actually doing.

WHEN GENES AREN’T ENOUGH

For decades, medicine has focused intensely on genes. The Human Genome Project promised that by reading DNA’s instruction manual, we could transform treatment for illnesses. And in many ways, it has. Genetic testing now identifies potential treatments for high-risk childhood cancers and rare diseases. But genes tell only part of the story.

Think of your **genome as an architect’s blueprint**. It shows what could be built but reveals nothing about what’s actually happening on the construction site. Genes provide instructions on how to put together **proteins—the physical building blocks and machinery of the cells**, whose activity changes constantly.



Some proteins swirl around cells in the thousands, actively transporting materials or transmitting signals. Others provide structural support. Unlike our relatively static genome, this protein landscape shifts throughout the day, responding to everything from exercise to medications to the first signs of illness.

PROTEIN PIONEERS

The challenge in studying proteins has long been technology. While reading DNA became routine, analyzing proteins remained too complex and expensive for widespread use. That’s changing rapidly, thanks to donor support that has brought cutting-edge equipment and technology to BC Children’s Hospital.

Dr. Lange’s team is the first in Canada to use state-of-the-art mass spectrometers for protein analysis to inform treatments of children with cancer. The machine works like an incredibly precise scale, weighing proteins and their components to create molecular fingerprints that reveal which cellular pathways are active, which are blocked and how tumours are adapting to treatment.

“Most cancer drugs actually target proteins, not genes,” Dr. Lange explained. “By measuring proteins directly, we can see which drugs will work for each specific child. It’s a second layer of insight, showing us how cancer cells are functioning in the moment.”

His team pioneered this approach in Canada, becoming the first to use protein analysis to guide treatment decisions in children with cancer. The results proved so successful that they’re now leading efforts in the Canadian Pediatric Cancer Consortium ACCESS to implement similar programs at The Hospital for Sick Children (SickKids) in Toronto and other hospitals across the country.

The potential became clear last year when they treated a young patient with cancer whose genetic testing offered no clear path forward. Protein analysis revealed an unexpected vulnerability in how the tumour cells produced energy. The solution? It turned out to be an antidepressant that happened to block this specific pathway to slow down the growth of the tumour. Without protein analysis, this connection would never have been made.

YOUR PERSONAL PROTEIN SIGNATURE

Successes like this hint at a revolutionary shift in how experts monitor health. Beyond guiding treatment decisions, proteins offer something even more powerful. They may help detect illness long before kids feel sick.

Unlike genes that, apart from cancer, remain largely unchanged throughout life, proteins shift and signal in real time. When conditions begin to take hold, proteins often sound the alarm months or even years before symptoms appear.

“Everyone has their own baseline that they tend to return to,” Dr. Lange said. “We’ll establish what a child’s proteins look like when they’re healthy and when they’re getting sick. Then we’ll spot dangerous changes much sooner.”

Artificial intelligence will help recognize these individual patterns, flagging unusual changes before symptoms appear. It’s precision health in its truest form, understanding disease as it unfolds in each unique person. “We’re returning to what medicine should be,” Dr. Lange added. “Technology is giving us the tools to see every patient as an individual, and to treat them that way.”

By examining both genes and proteins together, medicine is moving closer to delivering truly personalized care. The proteins in our bodies tell stories that genes alone can’t—stories of health and disease unfolding in real time, waiting for those with the tools to listen.★



Farah Jayousi, PhD student, prepares samples

A first among 8 billion

For a child with a rare disease, isolation and loneliness can run deeper than most people realize. They might be the only person in their country—sometimes one of just a handful in the world—living with their condition. No support groups exist, and no other families have walked this path before them. Few understand this heartbreaking reality better than Ebrahim.

From his earliest memories in Syria, Ebrahim faced mysterious health challenges. He struggled to move his left leg, elbows and hips as he grew. His immune system failed in ways that baffled everyone, leaving him to battle constant infections. After moving to Canada, Ebrahim began seeing an orthopedic surgeon at BC Children's Hospital. There, doctors discovered yet another challenge: cancer.

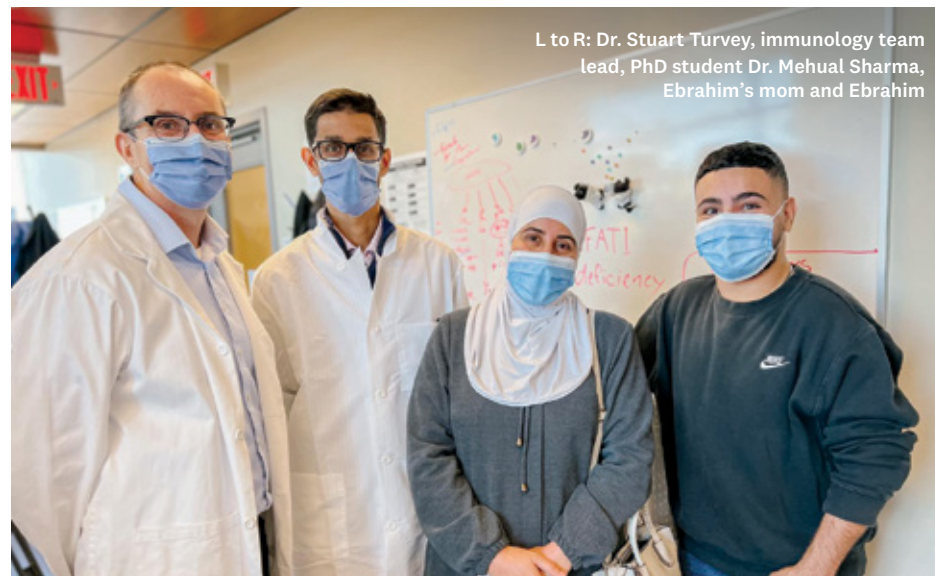
But his care team noticed something others might have missed. These scattered symptoms—the movement problems, the infections, now cancer—seemed connected. They suspected a single thread ran through everything. That hunch was confirmed through genetic testing. One altered gene lay at the root of everything.

To grasp the significance of this discovery, imagine DNA as an instruction manual the body uses to build proteins and function properly. A single typo—a mutation—in that manual can change how the body operates. For some, that's all it takes to trigger a rare disease.

Without extensive testing, Ebrahim couldn't be certain that this was the precise cause of his health challenges. Then in 2019, the Rare Disease Discovery Hub launched at BC Children's Hospital Research Institute. Ebrahim became its first patient—and shortly after, the first person in the world—diagnosed with complete NFAT1 deficiency. That single gene mutation disrupted his body's ability to produce

NFAT1, a protein essential for immune cell function and cartilage growth.

After a decade of questions, Ebrahim finally had a name for what was happening inside his body. While no cure exists yet, having a diagnosis changed everything. It meant the end of his diagnostic odyssey, and it opened the door to targeted treatment in the future.



L to R: Dr. Stuart Turvey, immunology team lead, PhD student Dr. Mehul Sharma, Ebrahim's mom and Ebrahim

BC Children's continues leading this work on a global scale. Their immunology team has discovered approximately 6% of all newly identified immunological disorders worldwide in the past two years—an extraordinary contribution to medicine. Each discovery creates a ripple effect. When the team publishes their findings, children around the world with the same undiagnosed conditions

can finally get answers.

Ebrahim knows how important this work is to kids facing uncertainty. While he was the first to be diagnosed with his condition, he knows he won't be the last.

"I would be really happy to know that what I've gone through can help someone else get their answer faster," he said. "Maybe, we can even figure out a treatment."*



Ebrahim as a young boy



Life's dimmer switch

How lived experience rewrites biology

A simple cheek swab from a four-year-old can reveal something surprising. Biologically, that child might be three years old. Or seven. Their chronological age and biological age don't match—and the difference tells a story about what they have experienced.

This is the power of the pediatric epigenetic clock developed by Dr. Michael Kobor, an investigator at BC Children's Hospital Research Institute. Now used in more than 80 research collaborations around the world, the tool measures something previous generations of doctors

couldn't see: how lived experience shapes biology at the molecular level.

"We have 25,000 genes," Dr. Kobor explained. "Imagine all of them like light bulbs. If you were to take a microscope and look into your cells, you would see 25,000 different light bulbs lighting up to various degrees. What sets the intensity of those light bulbs is a dimmer. That dimmer is epigenetics."

As it turns out, those dimmers can respond to everything. When Dr. Kobor's team studied children with obsessive compulsive disorder, they discovered accelerated epigenetic aging. Now, they're studying whether therapy might alter those changes. The clock also revealed how early birth affects biological timing, and how interventions might change the epigenetic aging process in kids who have faced adversity.

One of the most ambitious efforts in Dr. Kobor's lab is the BC Wildfire Study, generously funded by Genome BC, Canadian Institutes of Health Research and donors through BC Children's Hospital Foundation. The project will follow the natural experiences of 1,200 children across the province to answer pressing questions: What happens when infants are exposed to wildfire disasters in their earliest days, including before birth? How do cascading stresses—families fleeing, homes lost, food access disrupted—become biologically embedded?

"There's plenty of evidence that early environments get under the skin to affect health," Dr. Kobor said. But these changes aren't set in stone. The epigenetic clock offers a way to measure those impacts—often before they show up in traditional health assessments. And what can be measured can potentially be altered.

"In precision health, it's essential to take a holistic view," he added. "We don't just look at blood pressure—we consider the full lived experience and environment of a child so we can tailor care to their unique needs."*



Dr. Bruce Carleton with technology used to detect and quantify DNA in genomic research

The right match

Tailoring medications to each child's unique genetic code

In the early 1990s, Dr. Bruce Carleton, an investigator at BC Children's Hospital Research Institute, began observing something that puzzled clinicians everywhere. Children with the same cancer diagnoses, receiving the same drugs at identical doses, experienced vastly different outcomes. While some thrived, others developed serious, sometimes life-threatening, side effects.

The chemotherapy that effectively treats one child might cause permanent hearing loss in another. A medication that helps one patient could trigger heart complications in the next. At the time, there seemed to be no way to predict who would fall into which category.

Then came a breakthrough in the early 2000s that would change everything. After 13 years of effort, scientists

released a comprehensive map of the human genome—and Dr. Carleton saw an opportunity. “The human genome is a window into the biology that underlies these pathways of harm,” he explained. “I started thinking about how we could use this to improve the safety and efficacy of prescription drugs.”

THE COST OF ONE-SIZE-FITS-ALL

Dr. Carleton's research focuses on pharmacogenomics. The concept is straightforward: by understanding a patient's genetic profile, clinicians could better predict how they would respond to specific medications.

He concentrated initially on childhood cancer's most serious side effects. Seventy per cent of children with cancer receive drugs that can

cause heart complications. Another commonly used chemotherapy drug causes permanent hearing loss in some kids. The standard approach—monitor and adjust after problems develop—often means interventions come too late.

“If we're treating cancer, let's deal with that—not create another problem we have to manage along the way,” Dr. Carleton said. “If you can predict, you can hopefully prevent.”

FROM LAB TO BEDSIDE

With donor support powering this research since day one, Dr. Carleton has studied the genetic profiles of over 12,350 kids—including 1,000 across Canada last year alone. Today at BC Children's, nearly every child

“We're moving from ‘you have lymphoblastic leukemia’ to ‘you are Mary and you have this cancer and this is how we're going to treat your leukemia.’ We need to celebrate those differences and come up with tools that allow us to do that. That's what kids deserve.”

— Dr. Bruce Carleton

receiving cancer treatment undergoes genetic testing. This information allows clinicians to adjust treatment plans before the first dose. That might mean modifying dosages or choosing different medications based on each child's unique genetic profile.

The impact has been remarkable. Countless children have avoided serious complications that would have shadowed them for life. Now, the approach is expanding beyond oncology.

“I'm seeing increased requests from community pediatricians as well as areas like mental health to provide pharmacogenomic test results,” Dr. Carleton noted. In psychiatry, genetic testing can help identify which antidepressant is most likely to be effective for a particular child, reducing the often lengthy trial-and-error period that families endure.

PRECISION PAIN MANAGEMENT

The foundation Dr. Carleton built over the decades is now enabling the next generation of researchers to tackle challenges just as pressing. Dr. Catrina Loucks, a researcher also studying pharmacogenomics, is addressing a delicate balance in pediatric pain management. While opioid safety is a valid concern, it has inadvertently led to under-treatment of pain in some children, including those with cancer who often experience highly-distressing pain.

“Undertreated pain can impair brain development, disrupt behaviour and increase the risk of chronic pain later in life,” Dr. Loucks explained. “Instead of completely avoiding opioids, our work is trying to use precision health strategies to help identify children who are most likely to benefit without harm.”

Her team analyzes treatment data from thousands of children over two decades, searching for genetic variants that predict opioid responses. Next, they assess genetic variants in a worm model to explore how they impact opioid responses. The goal is to optimize pain management while minimizing risks—so that kids get the relief they need without unnecessary dangers.

MANAGING THE DATA CHALLENGE

Success has created new complexities, especially around data management. Each whole genome sequence generates 100-150 terabytes of information. Comparing hundreds or even thousands of patients requires analyzing massive datasets to identify meaningful patterns—a task that would have been impossible just years ago.

“That's where AI is becoming helpful,” Dr. Carleton said. With donor support, his team is building systems that can analyze massive genetic datasets as intuitively as searching Google.

“I want to ask questions like: ‘How many patients had toxicity from this drug? Compare their genomes with those who didn't and tell me what gene variants are different,’” he said. “Or, ‘tell me which genes occur in at least 50 percent of patients with toxicity.’”

TOWARD INDIVIDUALIZED CARE

As Dr. Carleton explains, this represents a shift toward truly personalized medicine—to a future where medicine doesn't just treat diseases, but treats people.

“It's moving from ‘you have lymphoblastic leukemia’ to ‘you are Mary and you have this cancer and this is how we are going to treat your leukemia,’” Dr. Carleton said. “What are your risks and how are you different from others with the same cancer type and how might we treat you differently? We need to celebrate those differences and come up with tools that allow us to do that. That's what kids deserve.”✱



L to R: Yoonsoo Ham, undergraduate research trainee, and Dr. Catrina Loucks in the lab

The power of being seen

“I need treatment that’s not treatment.”

Dr. Danya Fast, a medical anthropologist at BC Children’s Hospital, hears this phrase often from youth in crisis. It may sound paradoxical, but it reveals a truth that medicine is only beginning to understand. For many young people facing mental health and substance use challenges, what brings them to the hospital isn’t simply their medical symptoms. Often, they’re also carrying the weight of housing instability, poverty or isolation.

“What we often hear from young people is that it’s impossible to get a true picture of their needs, priorities and desires without understanding that bigger picture,” explained Dr. Fast.

THE LIMITS OF TREATMENT ALONE

Consider a teenager in crisis with substance use disorder.



Clinicians can prescribe medications and provide counselling, but when that young person returns to unstable housing or crushing poverty—with no access to sources of joy, meaning or community—the underlying conditions driving their crisis can remain unchanged.

“Medications can address some of the symptoms, but we won’t actually be treating them with a view toward

long-term wellness,” she said. This creates a cycle where symptoms get treated while root causes persist.

A NEW APPROACH

Dr. Fast is leading groundbreaking work to change that pattern. Funded by the National Institutes of Health, and developed in collaboration with Harvard University and Boston Medical Center, she’s creating a youth overdose

prevention toolkit that reimagines what care can look like. Instead of focusing only on medical interventions, the toolkit guides health care providers to open conversations about housing, social supports and personal goals.

When clinicians ask about housing stability and social connections alongside medical symptoms, young people feel genuinely seen and heard—and they’re more likely to return for follow-up care. This approach also enables health care providers to connect patients with social workers and community resources that address underlying social determinants of health.

STARTING SMALL

Dr. Fast acknowledges the challenge’s scale: “When we talk about social and structural determinants of health, it can feel way too vast.” But she emphasizes that meaningful change can start in clinical encounters. “One thing we can do right away is educate clinicians on how important it is to open conversations with kids and families.”

Sometimes, being truly seen, heard and understood as a whole person navigating complex realities is exactly what youth need. In those moments of connection, healing begins with the simple recognition that their struggles extend far beyond what any medication alone can address.*



8th Floor, For Children We Care 愛兒住院部
Oncology/Hematology/
BMT Inpatient Unit,
BC Children’s Hospital
December 24, 2024, 7:16 PM

“I took this photo on Christmas Eve. Josie was recovering from the stem cell transplant she received earlier that month, to help fight her brain cancer. Her immune system was so weak she couldn’t have visitors, but her care team did everything they could to bring the holidays to her hospital room. They put up decorations, and helped her set out milk and cookies for Santa. Best of all, on Christmas day, they cleared two very special people for a visit: Josie’s dad and brother. They hadn’t seen Josie in over two weeks, and she missed them a lot. It wasn’t the same as being home, but it meant so much that we could be together.”

— Jessica, Josie’s mom



Join the League of

MIGHTY!

Become a monthly donor
at **LeagueofMighty.ca**